Hiromu Taketsugu

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ImIntheMiddle | Google Scholar

Research Interests

Physical AI, Human Motion Estimation and Prediction, Computer Vision, Autonomous Driving, Human-Robot Interaction

Education

Ph.D. **Toyota Technological Institute**, Information-Aided Technology — Apr. 2025 – Present
- Advisor: Prof. Norimichi Ukita (Intelligent Information Media Lab.)

M.Eng. **Toyota Technological Institute**, Electronics and Information Engineering — Apr. 2023 – Mar. 2025
- GPA: 3.55/4.00 **(Ranked 2nd out of 50 students)**
- Advisor: Prof. Norimichi Ukita (Intelligent Information Media Lab.)

B.Eng. **Toyota Technological Institute**, Electronics and Information Engineering — Apr. 2019 – Mar. 2023
- GPA: 3.50/4.00 **(Ranked 1st out of 83 students)**
- Minor in Mechanical Engineering
- Advisor: Prof. Norimichi Ukita (Intelligent Information Media Lab.)

Experience

JSPS Research Fellow (DC1), Japan Society for the Promotion of Science — Aichi, Japan, Apr. 2025 – Present
- Competitive fellowship with 14% acceptance rate
- Research on human-aware robot navigation

Teaching Assistant, Toyota Technological Institute — Aichi, Japan, Apr. 2023 – Present
- Face Recognition (Spring 2023)
- Artificial Intelligence & Deep Learning Experiments (Spring 2024, Spring 2025)
- Computer Graphics (Fall 2024, Fall 2025, Fall 2026)

Machine Learning Engineer Intern (Perception), Woven by Toyota — Tokyo, Japan, Jun. 2026 – Aug. 2026
- Advisor: Dr. Igor Vasiljevic

English Facilitator, Toyota Technological Institute — Aichi, Japan, Apr. 2024 – Mar. 2025
- Facilitator of the “Lunchtime Chatting” sessions

Teaching Assistant, The University of Tokyo — Remote, Apr. 2022 – Mar. 2025
- TA for GCI (Global Consumer Intelligence) machine learning course
- Course attended by over 30,000 students per semester

Science Writer, AI-SCHOLAR — Remote, Oct. 2023 – Sep. 2024
- Writing articles describing the latest machine learning research papers

Visiting Student, Toyota Technological Institute at Chicago — Chicago, US, Sep. 2023 – Dec. 2023
- Advisor: Prof. Matthew R. Walter (Robot Intelligence through Perception Lab.)

Research Internship, Toyota Central R&D Labs., Inc. — Aichi, Japan, Jan. 2022 – Mar. 2022
- Advisor: Dr. Yoshihiro Ohama

Engineer Internship, FUJI OOZX, Inc. — Shizuoka, Japan
Jan. 2020 – Feb. 2020

- Inspection of engine valves

Publications

Hexa-P: Physical Plausibility-aware Pre-training for Pedestrian Path Prediction — Nov. 2026
Hiromu Taketsugu, Shohei Nobuhara, Norimichi Ukita
Pattern Analysis and Applications

PRISM: Predictive Representation of Interaction Style and Motion for Social Robot Navigation — Sep. 2026
Bo-Han Chen, ***Hiromu Taketsugu***, Norimichi Ukita
European Conference on Computer Vision (ECCV) 2026 Workshop on Agent in World

Energy-Regularized Imitation Learning for Force- and Work-Aware Robotic Manipulation — Sep. 2026
Toshiki Otani, ***Hiromu Taketsugu***, Norimichi Ukita
European Conference on Computer Vision (ECCV) 2026
Workshop on Force-Grounded, Cross-View Articulated Manipulation

Hexa-P: Physical Plausibility-aware Pre-training for Pedestrian Path Prediction — Jun. 2026
Hiromu Taketsugu, Shohei Nobuhara, Norimichi Ukita
10th International Workshop on Symbolic-Neural Learning (SNL), Poster

Re-EmLoco: Recovering Embodied Locomotion for Degradation-Robust Pedestrian Path Prediction — Jun. 2026
Hiromu Taketsugu, Shohei Nobuhara, Norimichi Ukita
10th International Workshop on Symbolic-Neural Learning (SNL), Poster

MMCM: Multimodality-aware Metric using Clustering-based Modes for Probabilistic Human Motion Prediction — Mar. 2026
Kyotaro Tokoro, ***Hiromu Taketsugu***, Norimichi Ukita
*IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**)*

Hierarchical Group-aware Token Merging for Efficient Trajectory Prediction — Oct. 2025
Yuki Yoshida, Norimichi Ukita, ***Hiromu Taketsugu***
9th International Workshop on Symbolic-Neural Learning (SNL)

Multi-Person Pose Estimation Evaluation Using Optimal Transportation and Improved Pose Matching — Jul. 2025
Takato Moriki, ***Hiromu Taketsugu***, Norimichi Ukita
*19th International Conference on Machine Vision and Applications (MVA), **Best Poster Award***

Selective Social-Interaction via Individual Importance for Fast Human Trajectory Prediction — Jul. 2025
Yota Urano, ***Hiromu Taketsugu***, Norimichi Ukita
The 28th Meeting on Image Recognition and Understanding (MIRU)

Physical Plausibility-aware Trajectory Prediction via Locomotion Embodiment — Jun. 2025
Hiromu Taketsugu, Takeru Oba, Takahiro Maeda, Shohei Nobuhara, Norimichi Ukita
*IEEE/CVF Conference on Computer Vision and Pattern Recognition (**CVPR**)*

Human Motion Prediction via Test-domain-aware Adaptation with Easily-available Human Motions Estimated from Videos — Apr. 2025
Katsuki Shimbo, ***Hiromu Taketsugu***, Norimichi Ukita

arXiv preprint arXiv:2505.07301

Active Transfer Learning for Efficient Video-Specific Human Pose Estimation Jan. 2024

Hiromu Taketsugu, Norimichi Ukita

*IEEE/CVF Winter Conference on Applications of Computer Vision (**WACV**)*

Uncertainty Criteria in Active Transfer Learning for Efficient Video-Specific Human Pose Estimation Aug. 2023

Hiromu Taketsugu, Norimichi Ukita

*18th International Conference on Machine Vision Applications (MVA), **Oral***

Domestic Presentations

Titles are English translations of the original Japanese titles.

Improving the Energy Efficiency of Robot Motion with an Energy Predictor Sep. 2026

Toshiki Otani, ***Hiromu Taketsugu***, Norimichi Ukita

Intelligent Mechatronics Workshop

2D Human Pose Estimation Handling Joints Inside and Outside the Image by Adaptively Integrating 3D Human Body Structure and Local Image Features May 2026

Takato Moriki, ***Hiromu Taketsugu***, Norimichi Ukita

IPSJ SIG on Computer Vision and Image Media (CVIM)

Efficient Trajectory Prediction by Token Merging with Hierarchical Group Structures May 2026

Yuki Yoshida, ***Hiromu Taketsugu***, Norimichi Ukita

IPSJ SIG on Computer Vision and Image Media (CVIM)

Physically Plausible Pedestrian Trajectory Prediction via Large-scale Pre-training Based on Physics Simulation Mar. 2026

Hiromu Taketsugu, Shohei Nobuhara, Norimichi Ukita

*245th IPSJ SIG on Computer Vision and Image Media (CVIM) Meeting, **Encouragement Award***

Evaluation Metric for Multi-Person Pose Estimation Using Optimal Transport and Region Partitioning May 2025

Takato Moriki, Takahiro Maeda, ***Hiromu Taketsugu***, Takeru Oba, Norimichi Ukita

IPSJ SIG on Computer Vision and Image Media (CVIM)

Stabilizing the Training of Walking Trajectory Prediction Consistent with Embodied Locomotion Nov. 2024

Hiromu Taketsugu, Shohei Nobuhara, Norimichi Ukita

IPSJ SIG on Computer Vision and Image Media (CVIM)

Human Trajectory Prediction with a Pose-consistency Loss Using a Physics Simulator May 2024

Hiromu Taketsugu, Shohei Nobuhara, Norimichi Ukita

IEICE Technical Committee on Pattern Recognition and Media Understanding (PRMU)

Human Trajectory Prediction via Density-based Anchor Selection in Contrastive Learning with Action Labels May 2024

Yota Urano, ***Hiromu Taketsugu***, Takahiro Maeda, Norimichi Ukita

IPSJ SIG on Computer Vision and Image Media (CVIM)

Uncertainty Criteria in Active Transfer Learning for Efficient Video-Specific Human Pose Estimation May 2023

Hiromu Taketsugu, Norimichi Ukita

IPSJ SIG on Computer Vision and Image Media (CVIM)

Awards

International Conference Research Presentation Award, IEEE Nagoya Section, for the presentation at CVPR 2025 — Apr. 2026

Encouragement Award at the 245th IPSJ SIG on Computer Vision and Image Media (CVIM) Meeting — Mar. 2026

Presentation Award at the 15th Interaction between companies and PhDs in Nagoya University — Aug. 2025

Best Poster Award at the 19th International Conference on Machine Vision Applications (MVA) — Jul. 2025

Poster Award (3rd place) at Japan Science and Technology Agency (JST) Adopting Sustainable Partnerships for Innovative Research Ecosystem (ASPIRE) Kyoto Workshop — Jul. 2025

Outstanding Student Award, the Society of Instrument and Control Engineers (SICE) — Mar. 2023

Toyoda Foundation Scholarship Award, Toyota Technological Institute — Mar. 2023

English Excellence Award (2nd place), Toyota Technological Institute — Mar. 2023

Outstanding Presentation Award for Bachelor's Thesis, Toyota Technological Institute — Jun. 2022

Outstanding Graduates (top 1%) of the Global Consumer Intelligence (GCI) 2021 Winter, The University of Tokyo — Mar. 2022

1st place in Advanced Technology Development, Toyota Technological Institute — Dec. 2021

1st place in the free-theme division of the English Science and Technology Presentation Contest, Toyota Technological Institute — Nov. 2021

Fellowships/Scholarships

Toyota Technological Institute Ph.D Program Scholarship – Special Grant — Sep. 2026
- **JPY 1,200,000/year** financial support

Toyoaki Scholarship — 2026 – Present
- **JPY 720,000/year** grant scholarship from the Toyoaki Scholarship Foundation

JSPS Research Fellowship for Young Scientists (DC1) — Apr. 2025 – Mar. 2028
- **JPY 200,000/month** financial support + **JPY 600,000** annual research grant

Toyota Technological Institute Ph.D Program Scholarship — Apr. 2025 – Mar. 2028
- Exemption from **JPY 700,000/year** tuition + **JPY 260,000** admission fee

Integrated Master's and Doctoral Program — Apr. 2023 – Mar. 2028
- **JPY 80,000/month** financial support during the Master's program + Exemption from **JPY 260,000** admission fee

IISF Grants for Researchers Attending International Conferences — Jun. 2025
- **JPY 240,000** travel grant upon attendance at CVPR 2025

NEC C&C Foundation Grants for Researchers Attending International Conferences — Jan. 2024
- **JPY 200,000** travel grant upon attendance at WACV 2024

Tatsuro Toyoda Foundation Scholarship — Sep. 2023 – Dec. 2023
- **JPY 1,000,000+** financial support for TTIC exchange program

Toyoda Foundation Scholarship Apr. 2019 – Mar. 2023

- **JPY 75,000 – 300,000/semester** tuition exemption

Projects

ICCV 2023, 4th Workshop on Visual Perception for Navigation in Human Environments: The JackRabbot Human Motion Forecasting Challenge Jun. 2023 – Aug. 2023

- Participated in a competition on end-to-end Human Trajectory Forecasting as a mentor to two intern students and achieved 4th place.

Survey on Online Prediction Apr. 2025 – Jul. 2025

- Conducted a survey on Online Prediction as a group leader in the MIRU2025 Young Researchers' Program. The survey results have been opened as GitHub repository (Awesome-Online-Prediction).

Academic Service

Invited Talks:

- "Physical Plausibility-aware Trajectory Prediction via Locomotion Embodiment", at The 28th Meeting on Image Recognition and Understanding (MIRU), Jul., 2025
- "Human Trajectory Prediction for Social Navigation", at the tech meetup with Toyota Frontier Research Center, May, 2024
- "Computer Vision for Social Navigation", at NTT, Inc. Atsugi R&D Center, May, 2024

Reviewer: Computer Vision and Image Understanding (CVIU), IEEE Transactions on Pattern Analysis and Machine Intelligence (T-PAMI), The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2024, 2025, The IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) 2026, 2027, European Conference on Computer Vision (ECCV) 2026, Conference on Neural Information Processing Systems (NeurIPS) 2026, British Machine Vision Conference (BMVC) 2026
Technical Chair: Asian Conference on Computer Vision (ACCV) 2026
Workshop Organizer: HUFPEI Workshop (International Workshop on Human Understanding, Future Prediction, and Embodied Intelligence), ACCV 2026
Course Material: Experiments in Artificial Intelligence & Deep Learning, Toyota Technological Institute (2024, 2025)

Skills

Programming Languages: Python, C, HTML, CSS, MATLAB, GAS, JavaScript, Bash, SQL

Frameworks & Libraries: PyTorch, Numpy, Pandas, OpenCV, Hydra, Weights & Biases, Scikit-learn, and more

Tools: Linux (Ubuntu/WSL), Git, IsaacGym, LaTeX, VSCode, Docker, ROS, Vim, Unity, Conda, uv, and more

Languages: Japanese (Native), English (TOEFL iBT 82), Vietnamese (Intermediate, Duolingo 700+ day streak)

Certification:

- Fundamental Information Technology Engineer, Information-technology Promotion Agency (IPA), Japan
- Deep Learning for GENERAL, Japan Deep Learning Association (JDLA)
- Driver's license, Japan

Leadership & Activities

Captain, Spartan Race Club Ares Toyota Technological Institute
Apr. 2022 – Sep. 2023

- Led the biggest Spartan Race team in Japan.

Organizer, vista Remote
Sep. 2020 – Present

- Discord community for goal setting and mutual progress tracking.